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Chapter 7
Playing Accurately, Part I: Cards Exposed Situations

Exploitive play is a two-step process. The first step is the processing of reading our opponents' hands and strategies, which we discussed in Chapter 5. The second step is deciding on an action in light of the information from the first step. The process here is simply to calculate the EV of the various action options and select the one that has the highest expectation.

In calculating the EV against our opponent's distribution, we will consider each hand he can hold in turn and weight the EV values against his distribution. Since in each case we know his cards, and we are playing just a single hand against his known hand, we may be able to gain some insight by having the players simply turn their cards face up and examine the play from there. This may seem to be a frankly uninteresting game; in many cases it would be, with made hands betting while draws chase or fold depending on pot odds, as we saw in Chapter 4. But some interesting situations can arise where the correct play is neither obvious nor intuitive.

Example 7.1
Consider the following example:

The game is $30-$60 7-card stud.
Player X has: A♥ K♦ Q♥ J♠ T♣
Player Y has: A♠ T♠ 8♠ 5♠ 2♣

The pot contains $400. Player Y has only $120 in chips left; player X has him covered, How should the play go?

In considering this, we can look at each player's situation separately. For player X, the initial decision is between checking and betting. However, we prefer to characterize strategic options in a more complex manner that reflects the thought processes behind the selection. So instead of two options-checking and betting-X actually has five options:

Check with the intention of folding to a bet.	(check-fold)
Check with the intention of calling a bet.	(check-call)
Check with the intention of raising a bet.	(check-raise)
Bet with the Intention of calling a raise.	(bet-call)
Bet with the intention of folding to a raise.	(bet-fold)

X can select from among these options the one with the highest value. We present X's options in this manner because the consideration of future actions on the same street is an important part of playing well. Many errors can be prevented or mitigated by considering the action that will follow when taking an initial action.

We can immediately calculate the expected value of the fourth and fifth options. The expected value of betting if Y folds or calls is the same no matter what X's intentions were if Y raises, so we can ignore those cases. If Y raises, there is no more betting on future streets, and the expected value of the bet-call option is:

<X, bet-call> = (*p*(X wins)) (new pot size) - (cost of action on this street)

The pot is currently $400; if both players put their remaining $120 in on this street, the new pot will be $640. Y will make a flush on sixth street $^{8}/_{42}$ of the time, and when he misses, he will

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make a flush on the river $^{8}/_{41}$ of the rime. X cannot improve, so X's chance of winning the pot is:

p(X wins) = 1 - p(X loses)
p(X wins) = 1 – {p(Y wins 6th street) + [p(Y misses 6th street)][p(Y wins river)]}
p(X wins) = 1 - [($^{8}/_{42}$) + ($^{34}/_{42}$)($^{8}/_{41}$)]
p(*X* wins) = 65.16%

So the expected value of X's bet-call option is:

<X, bet-call> = p(X wins) (new pot size) - (cost of action)
<X, bet-call> = (0.6516)($640) - $120
<X, bet-call> ≈ $297

If X bets and folds to a raise, then his expected value is:

<X, bet-fold> = - $60

These are not the total EVs of adopting these strategies-however, the EVs of the cases where Y doesn't raise are identical, hence we can compare these directly. It's clear from this that X's strategy of bet-call has higher expectation than his strategy of bet-fold. We can therefore eliminate bet-fold from consideration as a possible strategy for X. This should be intuitive-after all, X has almost two-thirds of the pot! Folding in a fairly large pot when his opponent is merely drawing to a flush would be disastrous.

In a like manner, we can examine the three options that begin with a check. If Y checks behind, then these three options have identical EV. If Y bets, then we have the following:

<X, check-fold> = $0

If X check-raises and Y calls, then X's EV is the same as in the bet-call case where Y raised. And if X check-raises and Y folds, X's EV is even higher:

<X, check-raise | Y calls > ≈ S297
<X, check-raise | Y folds> = $460

From this, we can eliminate the check-fold option. X can make more money by substituting the check-raise option for the check-fold option anywhere that he would play the latter.

With simple analysis, we have eliminated three of X's possible strategies, leaving us with:

check-call
check-raise
bet-call

To go further, however, we must consider what Y will do if X checks or bets. Suppose that we take as our guess at Y's strategy that he will check behind if X checks, and call a bet if X bets. On sixth street, what will happen? If Y has made a flush, then all betting will end-X will no longer call any bets because he has no chance of winning the pot. If Y has not made a flush, the hand has simplified to the case we looked at in Chapter 4. X will bet and Y will call or fold based on pot odds-in this case, Y will call a bet.

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Using this information, we can calculate the expected values of the three options for X.
<X, check-call> = p(flush on 6th)($0) + p(flush on 7th)(-$60) + p(no flush)($460)
<X, check-call> = $(8/42)$($0) + $(8/41)(34/42)$(-$60) + {1 - [$8/42$ + $(8/41)(34/42)$]}($460)
<X, check-call> = $(8/41)(34/42)$(-$60) + $(1 - 8/42 - (8/41)(34/42))$($460)
<X, check-call> ≈ $290.24
<X, check-raise> ≈ $290.24

The equity of check-raising and check-calling are the same because if X does check, Y will check behind, such that X doesn't get to Follow through with his call or raise intentions.

<X, bet-call> = p(flush on 6th) (-$60) + p(flush on 7th)(-$120) + p(no flush) ($520)
<X, bet-call> = $(8/42)$($60) + $(8/41)(34/42)$(-$120) + {1 - [$8/42$ + $(8/41)(34/42)$]}($520)
<X, bet-call> = $(8/42)$($60) + $(8/41)(34/42)$(-$120) + {1 - [$8/42$ - $(8/41)(34/42)$]}($520)
<X, bet-call> ≈ $308.43

Based on this, then, X should bet.

We have two final confirmations to make. Recall that we guessed at Y's strategy of checking behind X's check and flat calling X's bet. If Y bets behind X's check, X can simply flat call Y's bet and get $308.43. So Y does better by checking behind. We can calculate the EV of Y raising X's bet as well. Recall that we previously found that X's EV from the bet-call option when Y raised was $297. This is lower than $308.43; hence Y should instead raise. X can still do no better than his bet-call.

So the play on fifth street goes (if both players maximize their EV"):

X bets, and Y raises, and X calls.

This may look a little odd, as the draw raises the made hand's bet. X (the made hand) bets and charges Y to draw-however, it is in turn correct for Y to raise, while a 2-1 dog to make his hand! This occurs because the stacks are limited -Y is all-in and can't be punished by more betting. Since he has odds to call on 5th street for the 6th street card alone, he might as well make sure that all the money goes in on 5th street, so that if he should make his flush on 6th street, he'll still get the last bet hum the straight. This is because his draw is ***open*** - *that* is, X knows when he is beaten. In cases where the draw is ***closed,*** X may still have to pay off on sixth if Y makes his flush.

Example 7.2

Next we have a different example with limited stacks. In this case we allow the starting stack sizes to vary and examine the effect of the initial stack sizes on the play.

The game is pot limit holdem, but with a special rule that only pot-sized bets (or allin bets if the player has less than a pot remaining) may be made. We call this the ***rigid pot limit game;*** it is substantially simpler than the full pot limit game (where players may bet any amount between a minimum bet and the pot).

Player X has: A♥ A♦
Player Y has: 8♣ 7♣
The flop is: 9♣ 6♣ *2*♦

(We'll ignore runner-runner full houses for the AA and runner-runner two pair for the 87s for the sake of discussion. Assume that on each street the 87 simply has 15 outs and either hits or docs

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not.)
We can immediately calculate Y's equity if the cards are simple dealt out:

<Y> = 1 - *p*(miss twice)
<Y>= 1 - $(^{30}/_{45})(^{29}/_{44})$
<Y> = 56.06%

The pot contains $100, Player X is first to act. How should the play go for different stack sizes?

Case 1: Small stacks.
Let's first assume the stacks are $50. In this situation, player Y is the favorite if we simply dealt the cards out-he has 56.06% equity with his straight-flush draw. If Player X bets, then Yearly Player Y will call, yielding an EV of:

<X, X bets-Y calls > = *p(*X wins) (new pot value) - (cost of bet)
<X, X bets-Y calls> = [1 - *p(*Y wins)](new pot value) - (cost of bet)
<X, X bets-Y calls> = [1 - 0.5606]($200) - $50
<X, X bets-Y calls> = (0.4394) ($200) - $50
<X, X bets-Y calls > = $37.88

This is the equity of all the money getting in on the flop, no matter who bets first. It should be clear that if Player X checks, then Player Y can guarantee that X's equity is no greater than this - number by simply betting. X will have odds to call, and this same equity will be achieved.

If the play went check-check on the flop and Y failed to make a straight or flush ($^{30}/_{45}$ of the time) X could then bet the turn. In that case, X would have $^{29}/_{44}$ chance of winning. Y still has a clear call, getting more than 3 to 1 from the pot. X's equity, then is:

<X, bet turn> = [p(Y misses flop)][p(X wins)(new pot value) - (cost of bet)]
<X, bet turn> = $(^{30}/_{45})[(^{29}/_{44})(200) - (50)]$
<X, bet turn> = $54.55

This expected value is higher for X than the expected value when both players were all-in on the flop. And if it went check-check on the flop and Y did make his hand on the turn, X would simply fold. So X would prefer to have the action go check-check on the flop.

However, Y knows this as well. Since Y can limit X's equity to $37.88 by betting, the action should go check-bet-call on the flop. Notice that Player X's equity in the pot based on the showdown value of his hand is actually ($100) (1 - 0.5606) = $43.94, so the post-flop betting here reduces his equity by more than $6.

Case 2: Medium Stacks.

Now let's assume the stacks are $400. Again, player Y has a small edge if the money goes all-in on the flop with 56.06% equity.

If X bets, then Y has three choices-folding, raising all-in to $400 or calling $100. If he folds, then X's equity is $100.

<X, bet; Y, fold> = $100

If he raises all-in, then X's equity is:

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<X, bet; Y raise > = *p*(X wins) (new pot value) - (cost of bet)
<X, bet; Y raise > = [1 – *p*(Y wins)](new pot value) - (cost of bet)
<X, bet; Y raise > = (1- 0.5606)($900) - ($400)
<X, bet; Y raise > = (0.4394)($900) - $400 = - $4.55
<X, bet; Y raise > = -$4.55

If Y calls, then there are two things can happen:

$^{15}/_{45}$ of the time Y hits and X loses $100 net.
<X, bet; Y call | Y hits> = $ -100

$^{30}/_{45}$ of the time Y misses and the game simplifies to a pot odds game:
<X, bet; Y call | Y misses > = *p*(X wins) (new pot value) - (cost of bet)
<X, bet; Y call | Y misses> = ($^{29}/_{44}$)($900) - $400
<X, bet; Y call | Y misses> = $193.18

At this point, X will bet $300 and Y will be forced to call with $^{15}/_{44}$ equity in the pot. X's EV in this case will be

<X> = [*p*(X wins)](X's net when winning) - [*p*(Y wins)](X's net when losing)
<X> = ($^{29}/_{44}$)($500) - ($^{15}/_{44}$)($400)
<X> = $193.18

so X's overall equity will be:

<X, bet; Y call> = ($^{15}/_{45}$)(-$100) + $^{30}/_{45}$ ($193.18)
<X, bet; Y call> = $95.45

We calculated these EVs in X's terms-since this is a two player game and no one else can claim any part of the pot, Y can seek to either maximize his own expected value or minimize X's, and these will have the same effect. So it's clear that if X does bet $100, Y should raise all-in as this is the best of these three options.

Alternatively, X could check. If X checks, then Y can check or bet.

If Y checks, then again we have two outcomes. $^{15}/_{45}$ of the time Y hits and X nets $0. $^{30}/_{45}$ of the time, Y misses, and X bets the pot and is called.

X's overall EV, then is:

<X check; Y check> = [*p*(Y wins) (pot value to X)] + [*p*(X wins) (pot value to X-cost of bet)]
<X, check; Y check> = ($^{15}/_{45}$)($0) + ($^{30}/_{45}$)[($^{29}/_{44}$)($300 - $100)]
<X, check; Y check> = $65.15

Alternatively, Y can bet. If X raises all-in, he has -$4.55 again, while if he calls, he again makes $95.45.

Summarizing these outcomes (EVs from X's perspective):

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	Y's action				
X's action	**Check**	**Bet**	**Call**	**Raise**	**Fold**
Bet			$95.45	-$4.55	**$100**
Check	$65.15				
Check-raise	$65.15	-$4.55			
Check-call	$65.15	$95.45			

From this table, we can see that if X checks, the worst he can do is +$65.15 (Y checks behind,) So at this stack size ($400), the action should actually go check-check on the flop, Neither player can bet; if X bets, Y simply raises all-in and takes his edge on all the money, while if Y bets. X calls and gets the majority of the money in on the turn when he has a significant edge, Also notice that as the stacks have grown, X has gained because of the betting. In the small snack case, X actually lost money as a result of the betting (versus his showdown equity in the initial pot.) But here, X gains $65.15 - $43.94, or $21.19 from the postflop betting.

***Case* 3: Deep Stacks**
The last rigid pot-limit case we'll consider is stacks of $1300, or three pot-sized raises. This time, let's subdivide the EV calculations into four subcases:

Subcase a). *No money goes in on the flop.*
Since neither player will raise on the turn, X's equity in this subcase is the same as the receding case when no money went in on the flop, or $65.15

Subcase b). *One pot-sized bet ($100) goes in on the flop.*
This subcase, too, is similar to the preceding with X's equity at $95.45.

Subcase c). *Two pot-sized bets ($400) go in on the flop.*
In this subcase, X's equity is -$400 in the case where Y hits on the turn. When Y misses, X bets $900 and Y calls. $^{29}/_{44}$ of the time, X wins the pot and the rest of the time Y does.

<X, 2betsonflop > = ($^{15}/_{45}$)(-$400) + ($^{30}/_{45}$) [($2700)($^{29}/_{44}$) - $1300)]
<X, 2betsonflop > = $346.21

Subcase d). *Three pot-sized bets ($1300) go in on the flop.*
In this subcase, A simply gets his showdown equity from the pot:

<X. 3 bets on flop > = (1 - 0.5606) ($2700) - $1300
<X, 3 bets on flop > = (0.4394) ($2700) - $1300
<X, 3 bets on flop > = -$113.62

We can formulate some logical rules for the players' play based on these equities:

1) ***X will never put in the second bet on the flop.***
   If he does, Y will raise all-in. obtaining subcase d), the worst outcome for X.

2) ***Y will never put in the second bet on the flop.***
   If he does, X will flat call and we have subcase c), the worst outcome for Y.

3) ***X prefers subcase b) to subcase a) –***
   If he has a choice between zero or one bets, he will choose one.

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4) ***Y prefers subcase a) to subcase b) –***
If he has a choice between one and zero bets, he will choose zero.

From these rules, we can obtain the strategies for X and Y. Neither player will put in the second bet. Hence, both players have the option to put in the first bet if they wish. Since X prefers this option, he will bet, and Y will call. This results in an EV for X of $95.45. It is worth noting that X's expected value from betting if Y folded is just $100. In this case X gains a lot of value from the postflop betting-in fact the value of his hand has nearly doubled from $43.94 to $95.45.

When a made hand competes with a good draw, the draw generally does best by getting all the money into the pot as early in the hand as possible, while there are many cards to come. The made hand, by contrast, only wants to put in enough action such that he still has bets with which to punish the draw when the draw misses on the next street. In the case where where were only two bets left, if the made hand bet, the draw could get all-in. Instead, it's preferable for the made hand to delay betting so that he can extract value on the turn when the draw misses. But when three bets are left, the made hand can afford to bet the flop, knowing that the draw cannot prevent him from making a pot-sized bet on the turn.

Change the hands slightly by reducing player Y's outs by one, to 14, and now if there are three pot-sized bets left, player Y has to fold despite being the favorite to win the hand. Not only is there a pot-sized bet on the turn ($^{14}/_{44}$ being less than $^{15}/_{45}$), but also a pot-sized bet on the flop.

To this point, we've considered only the rigid pot-limit case, where the only bet size allowed is a pot-sized one. But in real pot-limit, either player can bet any amount up to the pot-how does this change the expected equities?

To examine this case, let's go back to the case where both players have two pot-sized bets remaining.

Example 7.3

The game is pot-limit holdem.
Player X has: A♥ A♦
Player Y has: 8♣ 7♣
The flop is: 9♣ 6♣ ***2♠***

(Again, we'll ignore runner-runner full houses for the AA and runnier-runner two pair for the 87s for the sake of. discussion-assume that on each, street the 87 simply has 15 outs.)

The pot contains $100, and both players have $400 in front of them.

We've seen previously that X's EV from betting $100 is -$4.55, while his EV from betting $0. (checking) is $65.15.

What if we allow X to bet any amount between those values? For example, suppose X bets $5.

We know that the action on the turn is deterministic-if Y hits on the turn, there will be no more betting, while if Y misses, X will bet either the pot or all-in. As a result, we should be able to express X's overall EV as a function of how much money goes into the pot on the flop total). Let x be the amount of money per player that goes in on the flop (up to $100 - we will address the case where more than $100 goes in momentarily). The pot will be (2x+$100) on the turn.

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X's expectation is as follows:

Outcome	*p*(Outcome)	<X> from outcome
Y wins on turn	15/45	-x
Y wins on river	(30/45)(15/44)	-x – (2x+100)
X wins	1 – (15/45) – (30/45)(15/44)	3x + 200

<X, x on flop> = $(^{15}/_{45})(-x) + (^{30}/_{45})(^{15}/_{44})(-3x - 100) + (1 - (^{15}/_{45}) - (^{30}/_{45})(^{15}/_{44}))(3x + 200)$

<X, x on flop> = $(^{1}/_{3})(-x) + (^{2}/_{3})(^{15}/_{44})(-3x - 100) + (^{2}/_{3} - (^{2}/_{3})(^{15}/_{44}))(3x + 200)$

<X, x on flop> = $(^{1}/_{3})(-x) + (^{5}/_{22})(-3x - 100) + (^{2}/_{3} - ^{5}/_{22})(3x + 200)$

<X, x on flop> = (-x/3) – 15x/22 - 500/22 + $(^{29}/_{66})(3x + 200)$

<X, x on flop> = -x/3 - 15x/22 - 500/22 + 29x/22 + 2900/33
<X, x on flop> = -x/3- 1500/66 + 7x/11 + 5800/66
<X, x on flop> = 10x/33 + $65.15

We can check this by comparing our values from the last analysis; <X. $0> = $65.15, and also <X, $100> = $95.45. This expression for <X, $x> is only valid for values from $0 to $100. Above $100, X can no longer bet the entire pot on the turn because the stacks are too small. Therefore, for values of x from $100 to $400:

Outcome	*p*(Outcome)	<X> from outcome
Y wins on turn	15/45	-x
Y wins on river	(30/45)(15/44)	-400
X wins	1 – (15/45) – (30/45)(15/44)	500

<X, $x> = $(^{15}/_{45})(-x) + (^{30}/_{45})(^{15}/_{44})(-400) + (1 - ^{15}/_{45} - (^{30}/_{45})(15/44)(400)$
<X, $x>= $(^{1}/_{3})(-x) + (^{5}/_{22})(-400) + (^{2}/_{3} - ^{5}/_{22})(400)$
<X, $x> = (-x/3) + $(^{5}/_{22})(-400) + (^{2}/_{3} - ^{5}/_{22})(500)$
<X, $x> = -x/3 + $128.79

For x = $400, for example, we get -$4.55 from this expression.

So now we have a two-part function that is X's EV for x getting into the pot on the flop:

x on [$0, $100]: <X, $x> = 10x/33 + $65.15
so <X, $0> = $65.15 and <X, $100> = $95.45
x on [$100, $400]: <X, $x> = -x/3 + $128.79
so <X, $100> = $94.45 and <X, $400> = - $4.55

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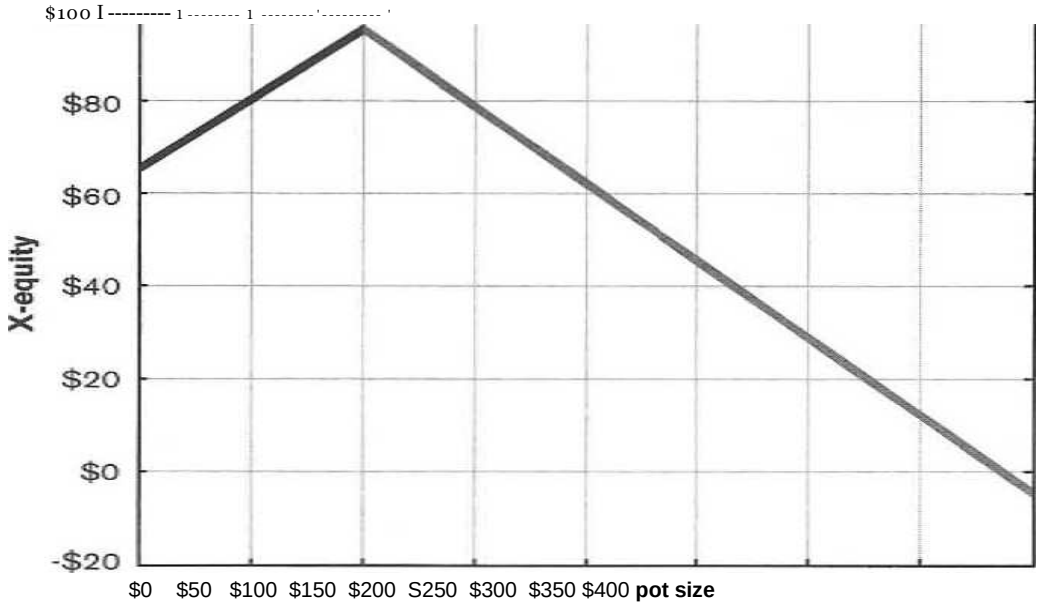

Figure 7.1. X's equity in Example 7.3 for various pot sizes

So let us again consider the case where X bets $5. Y has an infinite number of choices of raise amounts, but now that we can see the behavior of X's EV function, it is easy to discard many of the choices. Y can make it a total of $10 to $115, but if he chooses any value from $10 to $100, he simply adds to X's EV. Likewise, he can choose any amount from $100 to $115, but since all the additional money above $100 decreases X's EV, Y would choose to raise to $115 if he were going to raise an amount about $100.

So all that is left is to compare X's EV on these two actions:

<X, $x > = 10x/33 + $65.15
<X, Y calls $5 > = ($^{10}/_{33}$)($5) + $65.15
<X, Y calls $5 > = $66.67
<X, Y raises to $115 > = (- $^{1}/_{3}$)($115) + $128.79
<X, Y raises to $115 > = $90.46

Thus, Y will choose to simply call $5.

By betting $5, X has increased his equity by $1.50. Up to some point, for each dollar X bets he gains $^{10}/_{33}$ of a dollar in equity, If he bets $10, Y will again choose to call, but his equity is increased an additional $1.52 to $68.19. In fact, X can continue to bet larger and larger amounts until the values from Y's two options are equal. This will be the point at which X‘s expectation is maximum. When X bets the critical amount, Y has two options: to make the amount that goes into the pot $x, or to make a pot-sized raise and make the amount that goes into the pot (3x + 100).

<X, Y calls x> = <X, Y raises to (3x + 100)>
$^{10}/_{33}$ (x) + $65.15 = - 1/3(100+3x) + $128.79
$^{10}/_{33}$ (x) = - x + $30.30
x = $23.26

This is the proper amount For X to bet; it maximizes his expectation. No matter what action takes, X is guaranteed at least $72.20 in expectation:

<X, x = $23.26 on flop> = $^{10}/_{33}$ x + $65.15 = $72.20

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The upshot of all tills is that the stack sizes are of critical importance in determining how to play even when the cards are faceup. Additionally, as we can see from the results we obtained, deeper stacks favor the made hand because they allow the made hand to punish the draw on the turn when he misses. This runs somewhat counter to the usual wisdom about draws and made hands, which holds that draws benefit from deeper stacks and implied odds.

Another important concept from these examples is the following:

Good draws benefit a great deal from getting all the money in the pot early in the hand.

Example 7.4

Let's look at a common no-limit case.
No-limit holdem.
Player X has: A♥ K♦.
Player Y has: Q♥ Q♣.

It is preflop. Player Y posted a blind of $100. The stacks are $800.

Let's say that the hands are revealed sequentially, such that player X sees his AK first, acts, and then player Y reveals his queens. Assume that player X raised to $300 (an arbitrary but reasonable amount). What should player Y's strategy be?

AK benefits from seeing all five cards- it will only pair a third of the time on the flop, but will win 43% over all against QQ. The deeper the stacks, then, the more that the AK should want to get all the money in and the QQ should try to see the flop and push the AK out after the Hop if no A or K flops, right?

Wrong.

First, let's look at the shallower stacks:
Y with QQ is more than a 57% favorite; if he raises to $800, player X will be forced to call, getting 3 to 2 on his money. Raising to some lesser amount will not be better, as X can simply reraise to $800 to guarantee his equity mere. Jamming costs Y $700.

<Y, jam> = (0.5717) ($800 + $800) - $700
<Y, jam> = $214.72

On the other hand, let's say that Y with QQ calls. Then there will be $600 in the pot. The flop will contain an ace or a king but no queen about 30% of the time. In these cases, X with the AK can simply jam the flop, and Y will fold. The remaining 70% of the time, Y with the QQ will jam and X will fold with the AK.

<Y, call> = *p*(A or K flops) (Y loses his call) + *p*(A or K doesn't flop) (pot size)
<Y, call> = (0.3) (-$200) + (0.7) ($400)
<Y, call> = $220

So Y does slightly better here by calling and seeing if X hits the flop, then jamming if the AK misses.

Increase the stacks to $1800, however, and now the tables are turned;

<Y, jam > = (*p*(Y wins)) (pot size) - cost of jam

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<Y, jam > = (0.5717) ($3600) - $1700
<Y, jam > = $358.12
<Y, call > = still $220

So Y does much better by jamming immediately.

This runs counter to many players' intuitions about how the QQ/AK matchup might play out - with deep stacks, the QQ wants to get all the money in because of its significant equity advantage, while with shallower stacks it prefers to push the AK out of the pot on favorable flops. This conclusion is a little artificial-here the QQ can fearlessly push all his chips in the pot because he knows he has the advantage. In real holdem, however, he must fear AA or KK, and so he can't play in quite this way. However, many players overvalue seeing the flop, forgetting that their hand is not a simple "coin flip," but a substantial favorite over AK.

Going back to our original hypothesis that the action on each street should go "the made hand bets, and the draw either calls or folds depending on whether it has pot odds," we've seen that there are a number of situations where this is not at all the case. In pot-limit, for example, the size of the remaining stacks has a profound impact on the proper strategy for each player, and this is true in limit as well. In no-limit we saw a case where the stack sizes were integral to deciding whether to jam or just call.

Despite the occasionally interesting situations that occur in games where cards are exposed, you are unlikely to happen upon such a game. We present these situations, however, as primer material for the cases to follow; all the hand-reading skills in the world are of no use if one cannot play accurately when the cards are face-up, and while principles are often valuable as rules of thumb, recognition of situations where play might deviate from what is intuitively apparent can be worth equity.

Key Concepts

- Even when playing with the cards face-up, counterintuitive situations can occur.
- When considering the last raise all-in, draws should consider being more aggressive than is indicated by simple pot odds analysis. Further betting cannot hurt them and the made hand cannot get away if the draw hits when the money is all in.
- Stack size is of critical importance in big bet poker; even with the cards face-up changing the stacks changes the nature of the play drastically. Situations can even occur where the favorite has to fold to a bet from the underdog because of the potential for betting on later streets.
- Good draws (those which have close to 50°/o equity in the pot) benefit greatly from getting all the money in on the flop. However, if they cannot get all the money in (or enough that they cannot be hurt by later betting), they prefer to get as little as possible.
- Made hands often want to carry through a full bet for a later street when playing against a draw because of the heavy value extraction that can occur when the draw misses.

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